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Neocortical and hippocampal glucose hypometabolism following neurotoxic lesions of the entorhinal and perirhinal cortices in the non-human primate as shown by PET

Implications for Alzheimer's disease

K. Meguro,[1,*] X. Blaizot,[1,2] Y. Kondoh,[1,3] C. Le Mestric,[1] J. C. Baron[1] and C. Chavoix[1]

[1]*INSERM U320 and* [2]*CEA LRA 10V, Centre Cyceron, University of Caen, France and* [3]*Department of Neurology, Research Institute of Brain and Blood Vessels, Akita, Japan*

Correspondence to: Dr J. C. Baron, INSERM U320, Centre Cyceron, BP 5229, 14074 Caen cedex, France. E-mail: baron@cyceron.fr

**Present address: Section of Neuropsychology, Tohoku University School of Medicine, Sendai, Japan*

Summary

Temporoparietal glucose hypometabolism, neuronal loss in the basal forebrain cholinergic structures and preferential accumulation of neurofibrillary tangles in the rhinal cortex (i.e. in the entorhinal and perirhinal cortices) are three early characteristics of Alzheimer's disease. Based on studies of the effects of neurotoxic lesions in baboons, we previously concluded that damage to the cholinergic structures plays, at best, a marginal role in the association neocortex hypometabolism of Alzheimer's disease. In the present study, we have assessed the remote metabolic effects of bilateral neurotoxic lesions of both entorhinal and perirhinal cortices. Using coronal PET coregistered with MRI, the cerebral metabolic rate for glucose (CMR_{glc}) was measured before surgery and sequentially for 2–3 months afterward (around days 30, 45 and 80). Compared with sham-operated baboons, the lesioned animals showed a significant and long-lasting CMR_{glc} decline in a small set of brain regions, especially in the inferior parietal, posterior temporal, posterior cingulate and associative occipital cortices, as well as in the posterior hippocampal region, all of which also exhibit glucose hypometabolism in Alzheimer's disease. Remarkably, the degree of CMR_{glc} decline in four of these regions significantly correlated with the severity of histologically determined damage in the rhinal cortex, strongly supporting the specificity of the observed metabolic effects. There were also differences between the metabolic pattern observed in the lesioned animals and that classically reported in Alzheimer's disease; for instance, the hypometabolism we found in the stratum has not been reported in early Alzheimer's disease, although this structure can be affected in late stages of the disease and has direct anatomical connections with the rhinal cortex. Nevertheless, this study shows for the first time that the temporoparietal and hippocampal hypometabolism found in Alzheimer's disease may partly result from neuroanatomical disconnection with the rhinal cortex. This, in turn, further strengthens the hypothesis that neuronal damage and dysfunction in the rhinal cortices play a major role in the expression of Alzheimer's disease.

Keywords: rhinal cortex; baboon; CMR_{glc}; [^{18}F]fluoro-2-deoxy-D-glucose

Abbreviations: ANOVA = analysis of variance; CMR_{glc} = cerebral metabolic rate for glucose; FDG = [^{18}F]fluoro-2-deoxy-D-deoxyglucose; MRA = magnetic resonance angiography; pAC = posterior edge of the anterior commissure; RH = animal which received a lesion in the rhinal cortex; SH = sham-operated animal

Introduction

PET studies have shown that the early stages of Alzheimer's disease are characterized by metabolic declines that initially affect the temporoparietal association neocortex and secondarily the prefrontal association neocortex (Frackowiak *et al*., 1981; Benson *et al*., 1983; Cutler *et al*., 1985; Minoshima *et al*., 1997). The degree of neocortical hypometabolism is correlated with the severity of cognitive impairment (Haxby, 1990; Desgranges *et al*., 1998).

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Significant, albeit mild hypometabolism in the parietal association cortex is already present in pre-symptomatic mutation carriers in familial Alzheimer's disease (Kennedy *et al*., 1995; Perani *et al*., 1997). Although this hypometabolism may reflect local alterations such as synaptic loss, neurofibrillary tangles, amyloid deposition and other molecular changes, additional mechanisms may be involved. One classic hypothesis is that the hypometabolism of Alzheimer's disease may, at least in part, result from remote effects of neuronal loss in the basal forebrain cholinergic structures (Geaney *et al*., 1990). However, our previous [^{18}F]fluoro-2-deoxy-D-glucose (FDG) PET studies in baboons with extensive excitotoxic lesion of these structures showed that even severe cholinergic deafferentation only marginally affects resting cortical metabolism (Le Mestric *et al*., 1998). An alternative hypothesis would involve the early damage to a brain area endowed with a crucial anatomical position in the cortical circuitry (Mielke *et al*., 1996), such as the rhinal cortex.

Located in the ventromedial part of the temporal lobe, the rhinal cortex, which comprises the entorhinal and perirhinal cortices, has numerous reciprocal connections with neocortical associative areas and limbic structures (Van Hoesen and Pandya, 1975; Insausti *et al*., 1987; Suzuki and Amaral, 1994*b*). Notably, the perirhinal (together with the parahippocampal cortex) provides the main corticocortical input to the entorhinal cortex, and the latter is the major input to the hippocampus. One functional counterpart of this strategic neuroanatomical position is the severe memory impairment that results from lesions of the rhinal cortex in monkeys (Gaffan and Murray, 1992; Meunier *et al*., 1993; Blaizot *et al*., 1997).

According to Braak and Braak (Braak and Braak, 1991) and Van Hoesen and collaborators (Van Hoesen *et al*., 1991), the rhinal cortex is the area most and earliest affected by neurofibrillary tangles in Alzheimer's disease, even before the hippocampus. Both the entorhinal and perirhinal show atrophic changes with MRI (Pearlson *et al*., 1992; Juottonen *et al*., 1998). Furthermore, atrophy of the hippocampal formation is correlated with temporoparietal hypoperfusion (Jobst *et al*., 1992) or cerebral metabolic rate for glucose (CMR_{glc}) decrease (Meguro *et al*., 1997; Yamaguchi *et al*., 1997). Damage to the rhinal cortex could, thus, play a key role not only in the memory impairment, but also in the cortical hypometabolism, in Alzheimer's disease. Recent evidence indicates that the rhinal cortex is also consistently affected in non-demented elderly subjects though less severely than in Alzheimer's disease (Delacourte *et al*., 1999; Price and Morris, 1999).

Few studies, however, have directly investigated the effects of rhinal cortex lesions on CMR_{glc}. Although all three published rat studies (Jorgensen and Wright, 1988; Kurumaji and McCulloch, 1990; Beck *et al*., 1996) reported either no change in CMR_{glc} or a slight reduction only (mostly in the hippocampal area), the damage was limited to the entorhinal cortex (in whole or in part) and only one group performed bilateral lesions (Jorgensen and Wright, 1988). In addition, except in the study by Kurumaji and McCulloch (Kurumaji and McCulloch, 1990), necrotic lesions were made, so that none of these experiments mimicked the situation in Alzheimer's disease where damage is bilateral and characterized by neuronal loss only. Finally, these findings in rats would be difficult to extrapolate to man.

The present study was therefore performed in non-human primates in whom bilateral excitotoxic lesions of both the entorhinal and perirhinal cortices were created. A sham-operated group was used as the control, and both the lesioned and sham-operated animals were subjected to serial measurements of CMR_{glc} for 2–3 months by means of high-resolution, coronal PET scanning coregistered with individual MRI. Preliminary reports of this work have been published in abstract form (Meguro *et al*., 1995; Blaizot *et al*., 1997).

Method

Subjects

For this study, we used eight young adult male *Papio anubis* baboons (10–15 kg) which were divided into two groups that received either bilateral neurotoxic lesions of the rhinal cortex (RH group, $n = 4$, named RH1–RH4) or sham-operations (SH group, $n = 4$, named SH1–SH4).

Surgical procedure

The protocol was approved both by the INSERM Bureau of Animal Experimentation and by the INSERM Neurosciences Board.

In this protocol, MRI was used to (i) preoperatively determine the stereotaxic coordinates of the target sites in reference to MRI-visible implants anchored on to the skull prior to MRI (Le Mestric *et al*., 1993), and (ii) postoperatively confirm the location of the lesions. For both MRI scanning and surgery, the head of the anaesthetized baboon was placed in a non-ferromagnetic stereotaxic frame with the animal in the sphinx position. Adequate head positioning was controlled by an antero-posterior skull X-ray before each procedure. Heart rate, arterial pressure, body temperature and end-tidal CO_2 were continuously monitored. The surgical procedure is described in detail elsewhere (Blaizot *et al*., 1999).

Determination of the stereotaxic coordinates of the target sites

Target sites within the rhinal cortex were individually determined with the help of MRI T_1-weighted images as well as 2D-time-of-flight magnetic resonance angiography (MRA), which was performed in all baboons except for RH1 and RH2 which were lesioned before this methodology was operational (Blaizot *et al*., 1999). The use of MRA allows one to take into account the stereotaxic coordinates of the major cerebral vessels in selecting the trajectories of the

needle descents so as to avoid potentially serious haemorrhage. Indeed, because the anterior wing of the sphenoid bone hangs over the rhinal cortex, the classical vertical stereotaxic approach could not be used in this investigation. We thus designed an oblique approach through the lateral temporal cortex, a brain region highly vascularized by the temporal arteries. Both MRI and MRA acquisitions were performed in the same session using a GE Signa 1.5 T scanner with a 5″ receive-only surface coil, the baboon being anaesthetized with ketamine-xylazine [3.2–0.32 mg/kg/40 min, intramuscularly (i.m.)] and ventilated with enflurane (0.5–1.5%) plus N_2O : O_2 (2 : 1 v/v).

Briefly, the rhinal cortex boundaries were located on the MRI T_1-weighted images of each baboon using knowledge from both the cytoarchitectonic features described previously in macaques (Amaral *et al*., 1987; Suzuki and Amaral, 1994*b*) and a comprehensive set of coronal histological slices of a normal baboon's brain previously prepared in our laboratory. Twenty-two target sites per side were determined, 11 each for the entorhinal and perirhinal cortices, with two sites per mm in the antero-posterior direction (one site per rhinal area). To avoid extra damage, the rostrocaudal and mediolateral ends of the rhinal cortex were not targeted. The coordinates of the vessels detected by the MRA within the surgical zone were transferred on to corresponding coronal MRI T_1-weighted images (Blaizot *et al*., 1999). An oblique needle track was then drawn for each predetermined target site, tilted 25° from the horizontal plane, and slightly displaced in the parallel direction if it passed through or too close to a vessel. Finally, the 3D coordinates of each target site were measured in reference to six skull-implants containing a paramagnetic agent, four landmarks (two medial and two lateral) being used for each surgical side (Le Mestric *et al*., 1998; Blaizot *et al*., 1999).

Surgery

The bilateral lesions of the rhinal cortex were created in two stages (right hemisphere first), with a 5–14-day interval, on animals anaesthetized with isoflurane (0.8–1.5%) and N_2O : O_2 (2 : 1 v/v) and under conditions of muscular relaxation (atracurium 0.75 mg/kg/h, i.v.). The four relevant skull-landmarks were exposed, the temporal muscle was incised and a square-shaped craniectomy (~4 cm^2) with opening of the dura was performed over the temporal fossa. Neurotoxic lesions were made by injecting 1.2–1.5 µl of ibotenic acid at pH 7.4 (12–15 µg/µl in phosphate-buffered saline) into each of the 22 predetermined target sites, using a 10 µl Hamilton syringe fitted with a 30-gauge needle. After each injection, the needle was left in place for at least 3 min to prevent backflow along the needle track. After all injections were completed, the dura was sutured, the skull window was replaced and the temporal muscle, connective tissue and skin were sutured. An antibiotic (Cefamandole 15 mg/kg, i.m.) was administered once a day for 3 days and the baboon was observed daily until total recovery. The sham-operated baboons underwent the same surgical procedure except that the needle was lowered to a point 3 mm above the target sites and no injection was performed.

In order to confirm the position of the lesions soon after surgery and to assess putative postoperative haemorrhage, an MRI was performed 1–11 days after each operation in each baboon using T_1- and T_2-weighted 3D volume scan (Le Mestric *et al*., 1993).

PET procedure

Serial PET scanning was performed in the coronal plane, using the same stereotaxic frame as that described for the surgical procedure, which ensured excellent intra- and inter-subject reproducibility (Chavoix and Baron, 1995). Each baboon underwent one pre-surgical (baseline) and three sequential post-surgical PET scans, around days 30 (range 25–32 days), 45 (41–45) and 80 (75–93) after the last operation, using our previously described procedure (Le Mestric *et al*., 1998).

PET scans were performed with a 4-ring (7-slice) LETI TTV03 device (intrinsic resolution 5.5 × 5.5 × 9 mm, *x*, *y*, *z*) with the baboon under conditions of muscular relaxation and light anaesthesia/analgesia (phencyclidine: 7 µg/kg/min, i.v.; N_2O : O_2, 2 : 1 v/v); the effects of this regimen on cerebral metabolism has been shown to be negligible (Fitch *et al*., 1978). The baboon's head was positioned in the field of view of the PET camera using both intra-cerebral and external landmarks identified on prior sagittal MRIs, as previously reported (Le Mestric *et al*., 1993). The whole brain was sampled at six coronal planes located from pAC + 24 mm anteriorly (with pAC corresponding to the posterior edge of the anterior commissure) to pAC – 36 mm posteriorly.

After transmission scans with a ^{68}Ga–^{68}Ge source, an i.v. bolus of 1.7–2.6 mCi (i.e. 62.9–96.2 Mbq) of >99% pure FDG was injected. During PET data acquisition, which lasted 60 min, 27 sequential arterial blood samples were obtained to determine the plasma FDG time-activity curve, and the mean plasma glucose content over the whole scanning period was calculated from five measurements. PET images were reconstructed with the same field of view as that used for MRI.

Parametric images of CMR_{glc} (mg/100 g/min) were obtained by applying the three-compartment autoradiographic model of Phelps and colleagues (Phelps *et al*., 1979) on to the 50–60 min PET frames on a pixel-by-pixel basis, using the rate constants previously determined in our laboratory under the same experimental and anaesthesia conditions in the same species (K_1^* = 0.139, K_2^* = 0.212, K_3^* = 0.1, K_4^* = 0.0097; Miyazawa *et al*., 1993) and a lumped constant of 0.42.

Analysis of the PET data

Nineteen brain regions were investigated. The regions of interest (either circular, triangular or rectangular depending on the shape of the studied area, and with surface areas

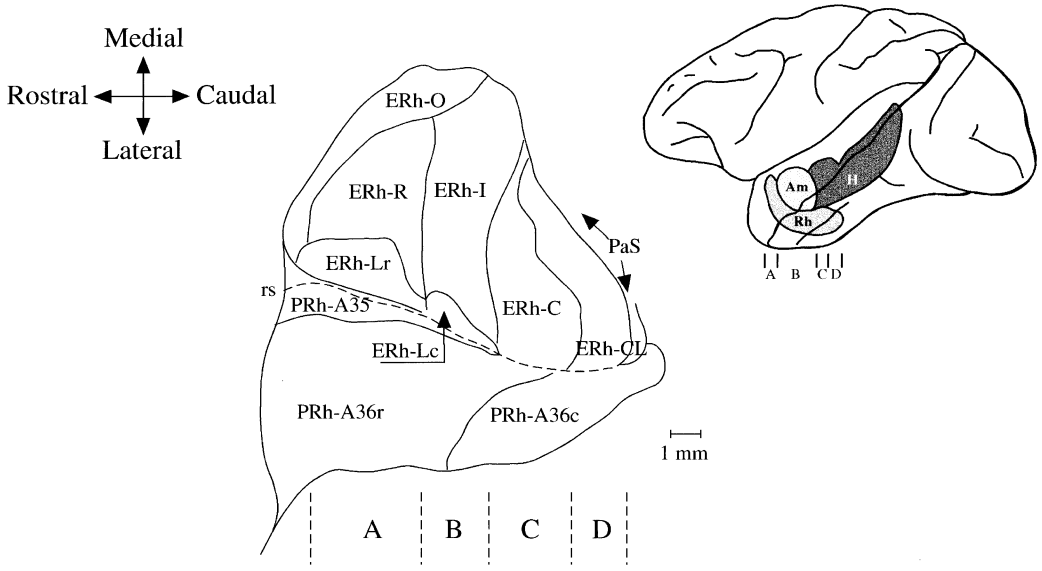


Fig. 1 Rhinal cortex subdivisions used for damage assessment. The rhinal cortex, shown here on an unfolded 2D map (modified from Amaral *et al*., 1987), was operationally divided into four antero-posterior segments, with reference to the adjacent amygdala and hippocampus (see insert), defined as levels A–D: level A, rostral part of the rhinal cortex to the anterior boundary of the amygdala; level B, from the latter to the amygdala-hippocampal junction; level C, running along the anterior half of the hippocampus; and level D, running along the posterior half of the hippocampus to the posterior boundary of the rhinal cortex. Thus, the rostral, intermediate, caudal and caudal limiting fields of the entorhinal subdivisions are mainly included in levels A–D, respectively, while Brodmann areas 35 and 36 of the perirhinal are mainly included in levels A and B, and C and D, respectively. Levels A–D represent area TE. Insert: levels A–D in relation to rhinal cortex, amygdala and hippocampus on a sagittal view of the brain; the relative width of the different levels differs between the two representations because of the unfolding in the main schema. Am = amygdala; ERh = entorhinal cortex; -C = caudal field; -CL = caudal limiting field; -I = intermediate field; -Lc = lateral field, caudal part; -Lr = lateral field, rostral part; -O = olfactory field; -R = rostral field; H = hippocampus; PaS = parasubiculum; PRh = perirhinal cortex; -A = Brodmann area; r = rostral portion; c = caudal portion; Rh = rhinal cortex; rs = rhinal sulcus.

~100 mm^2) were individually defined on six coregistered coronal MRIs with characteristics similar to the five PET slices (thickness, 9 mm; interslice, 3 mm). The regions of interest were then projected on to the corresponding CMR_{glc} images for calculation of regional CMR_{glc} as previously reported (Le Mestric *et al*., 1993). The following 18 regions of interest were defined on each side: primary and associative neocortical areas (n = 2 and 8, respectively), hippocampal regions (n = 2), basal ganglia (n = 3), thalamus, basal forebrain and cerebellar hemispheres. In addition, the anterior cingulate cortex was sampled with a single region of interest in the centre of both sides, placed on plane pAC + 24 mm (see Table 2 for a complete list and Le Mestric *et al*., 1998, for location of the bilateral regions of interest).

Post-mortem studies

At completion of the experiment (56–153 days post-surgery), the animal was deeply anaesthetized with enflurane (0.5–2%) and N_2O : O_2 (2 : 1 v/v), and perfused transcardially with cold heparinized saline. The brain was quickly removed and fixed in a 30% formalin solution for 24 h, then in 10% formalin for 1 week. A coronal block containing the entire temporal lobe was embedded in paraffin and cut into 15-μm coronal sections which were stained every 500 μm with thionin and every 1500 μm with GFAP (glial fibrillary acidic protein) antibody.

For semi-quantitative assessment of damage, the rhinal cortex was operationally divided into four antero-posterior segments defined as levels A–D (see Fig. 1). For each level, three to four brain sections were analysed histologically. The extent of damage was estimated semi-quantitatively on each section using microscopic examination for evidence of neuronal loss and gliosis by comparing the lesioned with the sham-operated brains. The neurons were counted within three microscopic fields per brain section analysed and per rhinal area. Because of the laminar organization of the anatomical connections (Insausti *et al*., 1987; Suzuki and Amaral, 1994*a*), the superficial layers (layers II and III) and the deep layers [layers IV (for the perirhinal only), V and VI] were separately analysed. For each subgroup of layers and within each

rhinal area, damage for a given level in each RH baboon was then calculated as the average percentage from the three to four sections included therein and for each side, as follows:

$$\%\ \text{of cell loss} = 100 - \frac{\text{number of neurons}}{\text{mean number of neurons in SH baboons}} \times 100$$

Similar histological assessment was performed for the hippocampus (levels C and D), amygdala (level B only) and area TE, the cytoarchitectonic field of von Bonin and Bailey (von Bonin and Bailey, 1947; levels A to D, all layers pooled), to evaluate potential extra damage.

Statistical analyses

Statistical analyses were performed on absolute CMR_{glc} values. Following averaging over both sides for the 18 bilateral regions of interest, inter-group differences at the baseline were first tested region-by-region by means of *t* tests. Then, global effects of the rhinal lesions were assessed using two-way analyses of variance (ANOVAs), with region and time (baseline, and days 30, 45 and 80 after surgery) as repeated measures. Region-by-region ANOVAs, with time as the repeated measure, were also performed. *Post hoc* analyses were computed with *t* tests.

Correlations between CMR_{glc} values for each region of interest on one hand, and histological damage in each rhinal area and the three adjacent structures on the other, were tested with Pearson's test across the four baboons of the RH group, but this time differentiating the two sides of the brain. Thus, for each region of interest, the correlation with damage in a particular structure was conducted on eight data pairs, two each per baboon (i.e. right-sided damage versus right-sided CMR_{glc} value, and left-sided damage versus left-sided CMR_{glc} value). The anterior cingulate region of interest was not assessed because a single region of interest covering both sides was used.

Results

Analysis of the lesion

Postoperative MRI

In both sides of all RH baboons except the left side of RH3, a clear-cut hypersignal was visible in the targeted areas on the T_2-weighted images. Regarding RH3, the hypersignal on the left side was found slightly dorsal, indicating a misplaced lesion, so that 10 additional sites were targeted on this side 14 days later. In RH2, both the T_1- and T_2-weighted images obtained 3 days after the left-sided lesion revealed an additional hypersignal in the Sylvian fissure, reflecting an intra-operative subarachnoid haemorrhage that prevented completion of the lesion (with only six injections out of the 22 planned being performed on that side). No signal changes were noted in the SH baboons (see Blaizot *et al*., 1999, for further detail).

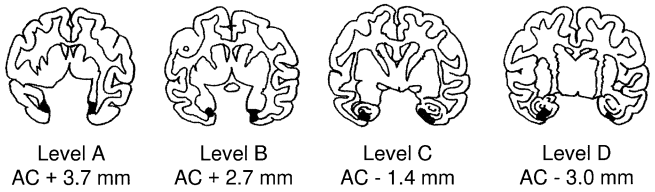


Fig. 2 Schematic damage in baboon RH4 illustrated on coronal sections at each of the four levels of the rhinal cortex, as defined in Fig. 1. All areas with a damage score higher than 25% (+ in Table 1) are blackened. Approximate rostrocaudal distance (in mm) from the anterior commissure (AC) is indicated below each coronal section.

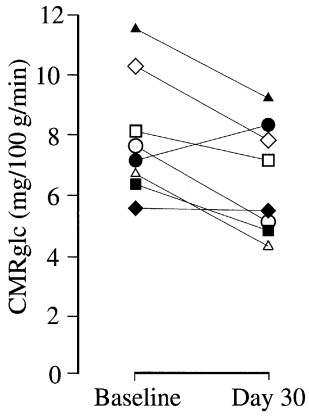


Fig. 3 CMR_{glc} (mean ± SD) at baseline and day 30 in the anterior temporal cortex, in the lesioned (RH, open symbols) and sham-operated (SH, solid symbols) baboons. The ANOVA showed a significant time effect but no group effect or interaction (see Results).

Histology

Table 1 shows a detailed analysis of the damage found in each animal. In all four RH baboons, each rhinal area (i.e. entorhinal and perirhinal) was damaged on both sides (levels B and C being the most consistently affected), with only mild and inconsistent extra damage. Within each rhinal area, the superficial and deep layers were comparably destroyed with, however, slightly more severe damage in the latter. Baboons RH1 and RH4 showed the best rhinal lesions, especially on the right and left sides, respectively, with minimal extra damage affecting only the hippocampus in RH4. In RH2, the distribution of damage on the left side reflected the lack of neurotoxic injections in the caudal part of the rhinal cortex due to intraoperative haemorrhage (see above). The lesional distribution is illustrated for one baboon in Fig. 2 (see Blaizot *et al*., 1999, for further detail).

In the SH group, only minimal gliosis was noted along the needle tracts, which passed through the hippocampus.

Table 1 *Distribution of damage in each of the four lesioned baboons (RH1–RH4)*

	Right side				Left side			
	A	B	C	D	A	B	C	D
RH1								
ERh superficial layer	+	+++	+++	++++	+−	+++	++	−
ERh deep layer	+++	+++	++++	++++	+−	+++	++	−
PRh superficial layer	++	+++	++++	++	++++	++++	+−	−
PRh deep layer	+++	+++	++++	++	++++	++++	+	−
Amygdala	na	+−*	na	na	na	+−*	na	na
Hippocampus	na	na	+*	+−*	na	na	+−*	−*
Area TE	−*	−*	+−*	−*	+−*	−*	−*	−*
RH2								
ERh superficial layer	−	+++	++++	++++	+++	++++	−	−
ERh deep layer	+	+++	++++	++++	+++	++++	−	−
PRh superficial layer	−	+	++++	++++	++++	+	−	−
PRh deep layer	+	++	++++	++++	++++	++	−	−
Amygdala	na	+*	na	na	na	−*	na	na
Hippocampus	na	na	++*	+−*	na	na	−*	−*
Area TE	−*	++*	−*	−*	−*	−*	−*	−*
RH3								
ERh superficial layer	−	−	++++	++++	++	++++	++++	++++
ERh deep layer	−	+−	++++	++++	++	++++	++++	++++
PRh superficial layer	−	++	++++	++++	+−	+	++	+++
PRh deep layer	+	++	++++	++++	+	+++	++++	++++
Amygdala	na	−*	na	na	na	+*	na	na
Hippocampus	na	na	+*	+*	na	na	++++*	+++*
Area TE	−*	+*	++*	+−*	−*	−*	−*	−*
RH4								
ERh superficial layer	+++	+++	+−	+−	++++	++++	++++	+
ERh deep layer	+++	++++	++++	+++	+++	++++	++++	+++
PRh superficial layer	++	++	+++	+	++++	++++	++++	++
PRh deep layer	+++	+++	++++	+++	++++	++++	++++	+++
Amygdala	na	−*	na	na	na	−*	na	na
Hippocampus	na	na	++*	+*	na	na	+−*	−*
Area TE	−*	−*	−*	−*	−*	−*	−*	−*

A, B, C and D represent the four anteroposterior levels passing through the rhinal cortex (ERh = entorhinal; PRh = perirhinal) as defined in Fig. 1. Damage for a given level is the mean damage across three to four histological sections. Semi-quantitative assessment was performed as explained in the Method section, and damage severity was expressed in percentage of cell loss, as follows: − = 0%; +− = <25%; + = 25–50%; ++ = 50–75%; +++ = >75%; ++++ = 100%; na = nonapplicable (absence of the structures at the level considered). *Non-targeted adjacent structures.

CMR_{glc} results

As expected, there was no significant difference in baseline CMR_{glc} values between the RH and SH groups for any of the regions investigated, except for the thalamus where the CMR_{glc} was slightly lower in the RH than in the SH group ($P < 0.05$, uncorrected for multiple comparisons).

For technical reasons, two post-surgical CMR_{glc} data sets were not available: day 45 for SH1 and day 80 for RH2. Therefore, the statistical analysis was performed on two different PET data arrangements: (i) all eight baboons, but two time points only (baseline and day 30; analysis I); and (ii) all four time points, but six baboons only (RH1, RH3, RH4 and SH2, SH3, SH4; analysis II).

The global ANOVA with two repeated measures (region and time) showed, for both analyses I and II, a significant region effect [$F(18,136) = 23.30$, $P < 0.001$ and $F(18,90) = 19.58$, $P < 0.001$, respectively] and a significant region × time interaction [$F(18,146) = 3.03$, $P < 0.001$ and $F(54,27) = 1.41$, $P = 0.04$, respectively]. In addition, analysis I revealed a significant time × group interaction [$F(1,6) = 7.72$, $P = 0.032$] and analysis II a significant region × group interaction [$F(18,72) = 2.94$, $P < 0.001$).

Concerning the region-by-region ANOVAs, analysis I showed a significant time effect ($P < 0.01$) for the anterior temporal cortex (a decrease from baseline in RH and SH baboons; Fig. 3) and a significant group × time interaction for the sensorimotor cortex ($P < 0.01$), the insula ($P < 0.03$), the anterior ($P < 0.03$) and posterior ($P < 0.001$) caudate nucleus, and the putamen ($P < 0.001$), due to a different CMR_{glc} evolution in the two groups from baseline

Table 2 *Regional CMR_{glc} values in the 19 brain regions in each group (*n = *3 per group; analysis II, see Results) at the four time points, and corresponding statistical findings*

Region	Group	Baseline mean (SD)	Day 30 mean (SD)	Day 45 mean (SD)	Day 80 mean (SD)	ANOVA*	*Post hoc t* tests†	
							Day 30	Day 4
Dorsolateral prefrontal	RH	8.4 (1.2)	7.6 (0.5)	7.0 (0.9)	7.2 (0.9)	n.s.	–	–
	SH	9.2 (2.6)	11.0 (3.1)	10.1 (2.8)	9.8 (1.3)			
Orbitolateral prefrontal	RH	9.1 (0.8)	7.2 (0.9)	6.9 (1.1)	7.6 (1.1)	n.s.	–	–
	SH	9.6 (2.6)	10.9 (3.1)	9.8 (3.4)	9.3 (1.3)			
Sensorimotor	RH	8.1 (1.3)	6.7 (1.0)	6.8 (0.8)	7.0 (1.0)	Group ($P = 0.04$)	0.01‡	0.04
	SH	8.8 (1.4)	9.9 (1.3)	9.2 (2.0)	9.8 (1.3)			
Inferior parietal	RH	7.4 (1.1)	6.1 (0.9)	5.6 (0.7)	6.3 (0.2)	Group ($P = 0.03$)	0.01‡	n.s.
	SH	8.4 (1.8)	9.4 (2.2)	8.0 (2.2)	8.2 (0.5)			
Anterior temporal	RH	7.4 (0.7)	5.5 (1.4)	5.3 (0.9)	5.5 (0.8)	n.s.	–	–
	SH	8.3 (2.7)	7.4 (2.1)	7.0 (1.2)	7.5 (2.1)			
Posterior temporal	RH	6.9 (1.2)	5.1 (1.6)	4.6 (0.9)	5.1 (0.3)	Group ($P = 0.03$)	0.01‡	n.s.
	SH	8.7 (2.3)	8.3 (1.6)	7.5 (1.2)	7.9 (1.1)			
Associative occipital	RH	6.2 (0.9)	5.5 (1.3)	4.6 (1.0)	4.9 (0.5)	Group ($P = 0.01$)	0.003‡	0.003†
	SH	7.1 (1.0)	8.6 (1.6)	7.8 (0.3)	8.4 (1.5)			
Primary occipital	RH	4.7 (1.1)	3.6 (0.4)	3.5 (0.3)	3.5 (0.4)	n.s.	–	–
	SH	4.7 (0.7)	5.5 (1.8)	5.1 (0.4)	5.4 (0.9)			
Anterior cingulate	RH	10.2 (0.8)	8.1 (1.2)	8.3 (1.0)	9.1 (0.4)	n.s.	–	–
	SH	10.1 (3.3)	11.8 (4.)	11.1 (3.6)	9.9 (2.1)			
Posterior cingulate	RH	7.6 (0.5)	7.0 (0.3)	5.8 (0.7)	6.4 (0.4)	Group ($P = 0.04$)	0.01‡	0.02
	SH	7.6 (0.9)	10.3 (3.0)	8.7 (1.9)	8.2 (0.6)			
Insula	RH	10.0 (1.2)	7.5 (1.9)	6.9 (0.8)	7.5 (1.1)	n.s.	–	–
	SH	9.7 (1.5)	10.1 (2.4)	9.4 (3.0)	10.1 (1.7)			
Anterior hippocampal	RH	5.1 (0.5)	4.1 (0.8)	4.0 (1.4)	3.6 (0.7)	n.s.	–	–
	SH	5.6 (1.2)	6.2 (1.7)	5.6 (0.9)	5.6 (1.1)			
Posterior hippocampal	RH	5.6 (0.9)	3.9 (0.2)	4.0 (0.3)	4.2 (0.5)	Group ($P = 0.04$)	0.005‡	n.s.
	SH	5.9 (1.3)	6.1 (1.3)	5.4 (0.5)	5.2 (0.8)			
Basal forebrain	RH	6.6 (0.7)	6.3 (0.4)	6.3 (1.5)	6.3 (1.5)	n.s.	–	–
	SH	6.3 (2.2)	7.3 (1.9)	5.7 (0.9)	6.2 (1.0)			
Anterior caudate (head)	RH	9.6 (1.5)	9.0 (1.1)	8.0 (0.4)	8.4 (0.9)	n.s.	–	–
	SH	7.4 (1.4)	9.5 (2.2)	8.6 (2.3)	8.5 (1.9)			
Posterior caudate (head)	RH	7.6 (0.4)	6.2 (0.6)	5.9 (1.4)	6.0 (1.4)	Group × Time ($P = 0.04$)	0.01	0.03
	SH	7.3 (1.3)	8.8 (1.5)	8.2 (1.0)	8.0 (1.2)			
Putamen	RH	9.8 (1.2)	8.0 (1.9)	7.9 (1.4)	7.9 (1.7)	Group × Time ($P = 0.03$)	0.02	n.s.
	SH	9.9 (0.6)	12.2 (2.6)	11.0 (3.0)	11.8 (1.9)			
Thalamus	RH	6.6 (0.4)	5.7 (0.6)	5.0 (1.2)	5.2 (0.8)	Group ($P = 0.01$)	0.001§	0.003‡
	SH	9.2 (2.9)	10.6 (1.9)	9.3 (0.7)	9.7 (1.4)			
Cerebellar lobe	RH	6.9 (1.5)	6.1 (0.2)	5.8 (0.5)	6.1 (0.3)	n.s.	–	–
	SH	5.8 (0.9)	7.2 (1.4)	6.6 (0.1)	6.9 (0.8)			

CMR values are given as mean (standard deviation). RH = rhinal group; SH = sham-operated group; n.s. = not significant. *Two-way ANOVA with time as repeated measure; †performed only if ANOVA was significant; ‡,§still significant after Bonferroni correction ($P < 0.05$ and 0.01, respectively).

to day 30, with a decrease in the RH group compared with a slight increase in the SH group. *Post hoc* tests were not significant for any region.

Concerning analysis II, Table 2 shows the mean ± SD CMR_{glc} data for all regions and both groups, together with the results from region-by-region ANOVAs, and *post hoc t* tests, including Bonferroni correction for multiple tests. It can be noted that the mean CMR_{glc} values were consistently lower in the RH than the SH group postoperatively, whatever the time point and region considered, whereas this occurred only at the level of chance preoperatively. The ANOVAs revealed significant effects in several regions. As with analysis I, there was a significant group × time interaction for the posterior caudate nucleus and the putamen, with *post hoc t* tests this time showing significantly reduced CMR_{glc} in the RH compared with the SH group at days 30 (both regions), 45 (posterior caudate) and 80 (putamen). In addition, a significant group effect was found for the sensorimotor, inferior parietal, posterior temporal, associative occipital and posterior cingulate cortices, as well as for the posterior hippocampal region and the thalamus. *Post hoc t* tests showed that the latter results were due to significantly reduced CMR_{glc} in the RH group (i) at all three time points for the sensorimotor and associative occipital cortices, and for the thalamus, and (ii) at one or two time points (consistently so at day 30) for the remaining four regions. These findings are illustrated in data plots for two selected regions and in CMR_{glc} images in Figs 4 and 5, respectively.

Rhinal damage–CMR_{glc} correlations

To test the correlations between histological damage and CMR_{glc} values in each region of interest, a damage score (average of the superficial and deep layers scores) was assigned to the perirhinal and entorhinal cortices as well as to the three adjacent structures (i.e. amygdala, hippocampus and area TE) for each side of the brain, from the detailed analysis shown in Table 1, with the scores –, +–, +, ++, +++, ++++ assigned values of 0–5, respectively. The correlations were tested for day 30 only, because this was the only post-surgical time for which PET data were available for all four RH baboons.

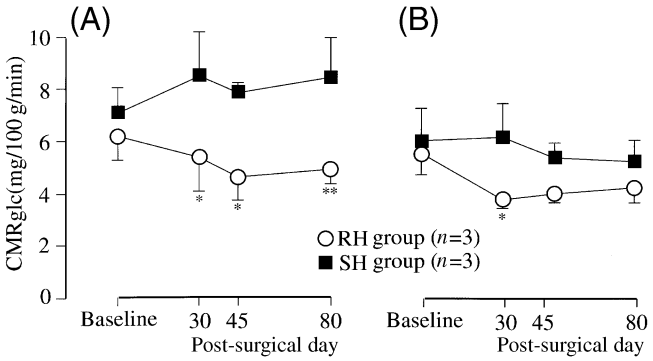


Fig. 4 Time-activity course of CMR_{glc} (mean ± SD) after induction of bilateral neurotoxic lesions of the rhinal cortex, from baseline to day 80, in the associative occipital cortex (**A**) and posterior hippocampal region (**B**). The CMR_{glc} values are the means of both sides. RH = rhinal group; SH = sham-operated group; $n = 3$ per group. The error bars are shown here only for purposes of illustration (the SD being used only for *post hoc* tests); * and **, $P < 0.005$ and 0.001, respectively, by *t* tests (still significant with Bonferroni correction; see Results section for findings with ANOVA).

Regarding the rhinal cortex, out of the 36 correlations tested (i.e. 18 regions of interest and two rhinal areas), nine (25%) were statistically significant ($P < 0.05$), all being in the neurobiologically expected direction (i.e. negative—the more severe the rhinal damage the lower the CMR_{glc}). Eight of them concerned damage to the perirhinal cortex, whose damage score correlated with CMR_{glc} in the dorsolateral prefrontal, sensorimotor, inferior parietal, associative occipital, primary occipital and posterior cingulate cortices, as well as with CMR_{glc} in the insula and the posterior hippocampal region (all $P < 0.05$, except sensorimotor cortex, $P < 0.01$). The last significant correlation concerned damage to the entorhinal cortex and CMR_{glc} in the orbitolateral prefrontal cortex ($P < 0.05$). Figure 6 illustrates typical correlations between damage score for each rhinal area and CMR_{glc} values for two regions.

In contrast, out of the 54 correlations tested, no significant correlation was found between damage scores for the three adjacent structures and CMR_{glc} for any region.

Discussion

This is the first report on the effects of a rhinal lesion on CMR_{glc} in non-human primates. We found that bilateral neurotoxic lesions of both entorhinal and perirhinal cortices

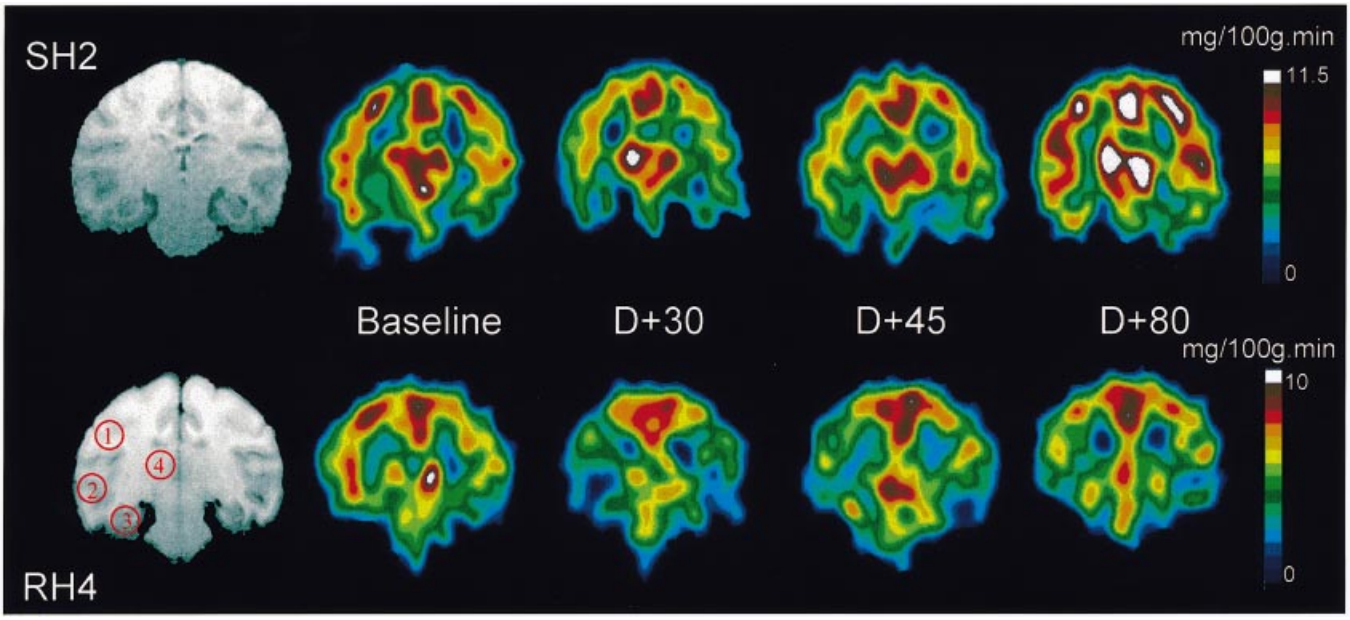


Fig. 5 Sequential coronal parametric PET images of CMR_{glc} in the pAC – 12 mm plane in a baboon with bilateral lesions of the rhinal cortex (RH4, bottom) compared with a sham-operated animal (SH2, top). These PET images were obtained before surgery (baseline) and at three post-surgical times. The pseudo-colour scale represents pixel CMR_{glc} values (mg/100g/min) as indicated, with the maximum value being adjusted to the range of values separately for each of the two baboons. The pAC – 12 mm plane is centred 12 mm caudal to the posterior edge of the anterior commissure, and the four regions of interest there (1, inferior parietal; 2, posterior temporal; 3, posterior hippocampal; 4, thalamus) are shown on one side of a corresponding MRI image (bottom left). Post-surgical decrease in CMR_{glc} can be seen in this RH baboon, especially in the inferior parietal, posterior temporal and posterior hippocampal regions.

induced a significant 30–40% decline in CMR_{glc} in several brain regions, which was long-lasting in a subset of them. Remarkably, the CMR_{glc} declines in both the hippocampal region and the association neocortex significantly correlated with the severity of histologically-determined rhinal cortex damage. Unexpectedly, however, no significant metabolic changes were found in the anterior hippocampal region of interest, which included part of the hippocampus and the amygdala, two regions highly connected with the rhinal cortex. Finally, a transient decline in CMR_{glc} characterized by a time effect without significant interaction, which indicates a non-specific effect in both RH and SH baboons, was observed in the anterior temporal cortex.

Before discussing these findings in terms of both implications for Alzheimer's disease and anatomical connections, we will first address some methodological issues.

The validity of our methodology for stereotaxic lesioning of the rhinal cortex has been extensively addressed in a previous report (Blaizot *et al*., 1999). Concerning the small samples inherent to studies in non-human primates, although it obviously hinders sensitivity, the use of both a longitudinal design and a correlational approach in this study largely compensated for this. Furthermore, we now have preliminary results from additional animals studied as part of another experimental protocol concerning the rhinal cortex, which not only confirm the present findings but enhance their statistical significance. In this study, true CMR_{glc} effects of the rhinal lesions may have actually been underestimated for two reasons. First, some metabolic recovery may have occurred during the necessary interval between right and left operations, especially so in the hippocampus as a result of sprouting by commissural/associational and septum fibres (Lynch *et al*., 1972; Matthews *et al*., 1976). Secondly, the rhinal damage, though extensive, was nevertheless incomplete, which may have resulted in submaximal effects. The damage distribution, however, was satisfactory, with essentially similar damage to both superficial and deep cell layers. Furthermore, there was only variable and very limited extra damage, which presumably did not substantially contribute to the metabolic changes, as suggested by the lack of significant correlation between damage to the three structures adjacent to the rhinal cortex and CMR_{glc} in the regions of interest assessed.

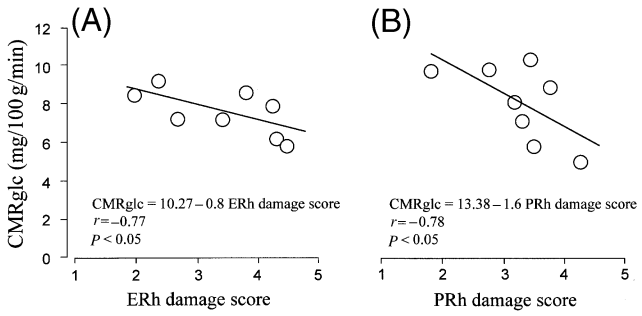


Fig. 6 Typical correlations between histological damage score and CMR_{glc}. (**A**) Entorhinal (ERh) cortex damage versus orbitolateral prefrontal cortex CMR_{glc}. (**B**) Perirhinal (PRh) cortex damage and insula CMR_{glc}. These correlations were obtained from data pairs ($n = 8$) from the four lesioned animals, with each point corresponding to a damage score on one side and CMR_{glc} on the same side.

Concerning CMR_{glc}, there was, superimposed on the above-mentioned significant effects in several regions, a more or less diffuse metabolic reduction, which most likely reflects large-scale transneuronal functional changes as a result of multiple focal alterations, as also occurs in multi-infarct dementia (Meguro *et al*., 1991) and late Alzheimer's disease (Cutler *et al*., 1985). Thus, severe damage to the rhinal cortex appears able to affect synaptic function across many brain regions. Regarding the regional CMR_{glc} effects, they can be classified as non-specific, specific or unexpectedly absent, and will be discussed in this sequence below.

Non-specific metabolic effects, in the form of transient CMR_{glc} declines in RH and SH baboons, were found in the anterior temporal cortex only, a region of interest that contained both superior and inferior temporal cortices, including area TE. This effect, which was significant from the baseline to the earliest time point only, is probably due to the needle passage in this area and represents a transient metabolic reduction as already reported in a related stereotaxic protocol (Le Mestric *et al*., 1998). These non-specific changes may, however, have masked a specific hypometabolism in the RH group, which would have been expected due to the strong neuroanatomical connections between the rhinal and the temporal cortices, and especially between perirhinal cortex and area TE (Webster *et al*., 1991). Indeed, an autoradiographic investigation in rats with entorhinal lesions did report specific, though transient, effects in the temporal cortex (Beck *et al*., 1996). Studies with longer postoperative observation may reveal this putative specific effect.

Specific effects of the rhinal lesions (defined here as group effects or group × time interactions with ANOVAs and/or as significant CMR_{glc}-damage correlations) were found in a relatively large number of regions. Remarkably, almost all those regions that showed consistently significant CMR_{glc} reductions in lesioned compared with sham-operated animals possess direct anatomical connections with the rhinal cortices, and have been reported to be preferentially hypometabolic in Alzheimer's disease (see below). As an independent statistical analysis, the finding of a high number of significant correlations between rhinal damage scores and CMR_{glc} values (9 out of 36 tests, or 25%), all in the neurobiologically expected direction, strongly supports the idea that we are indeed dealing with specific effects of the lesions. Furthermore, five of these nine significant correlations concerned brain regions that also showed significant ANOVA results. For the sake of clarity in the discussion below, these specific effects will be addressed according to three subgroups of regions, categorized with respect to both the findings with ANOVAs and correlations, and their relationships with Alzheimer's disease.

The first subgroup of regions comprises the inferior parietal,

associative occipital, posterior cingulate and posterior temporal cortices, as well as the posterior hippocampal area (including the hippocampus and parahippocampal cortex). These five regions show specific metabolic reduction and also exhibit glucose hypometabolism in Alzheimer's disease (Cutler *et al*., 1985; Kumar *et al*., 1991; Foster, 1994). All of them except the posterior temporal were remarkable in that they emerged as significant by both the ANOVA and the CMR_{glc} correlations with perirhinal damage. Regarding the posterior temporal region of interest, though it did not show significant correlations, it will also be discussed because it fulfilled the other two criteria (i.e. significant group effect with ANOVA and hypometabolism in Alzheimer's disease).

Previously, two autoradiographic rat studies reported a CMR_{glc} decrease in the hippocampus after uni- or bilateral ablation or electrolytic lesion of the entorhinal cortex (Jorgensen and Wright, 1988; Beck *et al*., 1996), with a long-lasting hypometabolism, up to 3 months in the latter. Beck and colleagues (Beck *et al*., 1996) also found a CMR_{glc} reduction in each of the other four regions of this subgroup, but metabolic recovery had supervened at 2 weeks post-surgery. In another study, in which unilateral ibotenic acid lesions were performed (Kurumaji and McCulloch, 1990), no CMR_{glc} changes were found, but the metabolic measurements were performed 14 days post-surgery and the lesions were limited to the caudal entorhinal cortex. Discrepancies between these and our study can be explained by the fact that unilateral lesions were induced in most of the rat studies, while in the only study where bilateral lesions were created, few regions were analysed. Furthermore, none of these previous rat studies targeted the perirhinal cortex, which, like the entorhinal cortex, has connections with the above five regions (see below). Finally, species differences may be argued.

In Alzheimer's disease, the posterior cingulate and the posterior parietal cortices are known to exhibit significant hypometabolism, from the very early stage of the disease, and to remain severely hypometabolic with disease progression (Minoshima *et al*., 1997; Desgranges *et al*., 1998). Hypometabolism in the association temporal and occipital cortices is found in mild Alzheimer's disease (Kumar *et al*., 1991; Pietrini *et al*., 1996). Discrepant findings have, however, been reported for the hippocampal region, as some authors found normal CMR_{glc} or blood flow (Duara *et al*., 1986; Fukuyama *et al*., 1991; Ishii *et al*., 1998), while others noted a decrease in glucose utilization (Grady *et al*., 1990; Kumar *et al*., 1991; Jagust *et al*., 1993), oxygen consumption (Ishii *et al*., 1996) or blood flow (Ohnishi *et al*., 1995; Julin *et al*., 1997). These discrepancies may be explained at least in part by the poor resolution of the PET devices used in most studies (Kumar *et al*., 1991; Minoshima *et al*., 1997) and the requirement of 3D PET data acquisition with special orientation relative to the long axis of the hippocampus for accurate analysis of this region (Julin *et al*., 1997).

Our findings regarding these five regions are also in agreement with their known anatomical connections with the rhinal cortex. In primates, both the entorhinal and perirhinal cortices are directly and strongly interconnected with the hippocampus and the parahippocampal cortex (Van Hoesen and Pandya, 1975; Suzuki and Amaral, 1990, 1994*a*, *b*), the two regions incorporated in our posterior hippocampal region of interest. In rhesus monkeys, severe and long-lasting (i.e. 18–28 months) transynaptic neuronal loss in the CA3 field has been reported after bilateral entorhinal ablation (Poduri *et al*., 1995), which would fit with the apparently permanent hypometabolism observed here. The rhinal cortex also has dense direct connections with both area TEO (Webster *et al*., 1991), which was included in our posterior temporal region of interest, and the posterior cingulate gyrus (Insausti *et al*., 1987). Regarding the inferior parietal and associative occipital cortices, connections with the rhinal cortex are mostly indirect, but this would not preclude the emergence of distant metabolic effects in these regions following rhinal damage. The fact that the significant correlations between CMR_{glc} reductions and rhinal damage concerned the perirhinal cortex only, whereas the four correlated regions (and especially the hippocampus) are mainly directly connected with the entorhinal cortex, does not exclude the hypothesis that damage to the latter also contributed, albeit to a lesser extent, to the observed hypometabolism. Furthermore, reciprocal connections exist between the perirhinal and entorhinal cortices (Suzuki and Amaral, 1994*a*).

The second subgroup of regions, comprising the dorsolateral and orbitolateral prefrontal cortices, were less consistently affected. However, these two prefrontal regions are usually hypometabolic in Alzheimer's disease in a later stage only (Frackowiak *et al*., 1981; Kumar *et al*., 1991; Foster, 1994). In the present study, their CMR_{glc} was correlated with the severity of damage in the perirhinal and entorhinal cortices, respectively, which is consistent with the notion that both prefrontal regions have connections with the rhinal cortex (Goldman-Rakic *et al*., 1984; Insausti *et al*., 1987). Finally, the lack of significant findings with the ANOVAs differs from the frontal hypometabolism induced by entorhinal lesions in rats (Jorgensen and Wright, 1988; Beck *et al*., 1996), but the latter effect was transient in this species.

A third subgroup of regions, comprising the thalamus, the striatum, the insula and the primary occipital and sensorimotor cortices, showed variably significant effects in the present study, but are relatively spared in Alzheimer's disease (Benson *et al*., 1983; Kumar *et al*., 1991; Foster, 1994). Autoradiographic studies in rats with entorhinal lesions also reported significant, though short-lasting, glucose hypometabolism in the occipital cortex and insula, but not in the thalamus and striatum (Jorgensen and Wright, 1988; Beck *et al*., 1996). Although our findings could be seen as only marginal, the significance in some of these seven regions is supported by the existence of direct anatomical connections with the rhinal cortex, especially between the perirhinal cortex and the caudate nucleus (Suzuki, 1996), the entorhinal cortex and the insula (Insausti *et al*., 1987), and between both rhinal areas and the thalamus (Russchen *et al*., 1987). The discrepancy

between our findings and the metabolic profile of Alzheimer's disease may be only relative, as these areas are indeed hypometabolic in moderate-to-severe Alzheimer's disease (Kumar *et al*., 1991), and especially so in subgroups with atypical metabolic patterns (Grady *et al*., 1990; Strassburger *et al*., 1996). In addition, the distribution of damage across the entorhinal and perirhinal cortices achieved in our study may differ from that occurring in Alzheimer's disease. For example, the preferential damage to layer II of the entorhinal cortex observed in the latter did not prevail in our baboons, although similar global neuronal loss was found in this rhinal area (slightly above 50% in our baboons compared with 32% and 69% in very mild and severe Alzheimer's disease, respectively; Gomez-Isla *et al*., 1996).

Finally, in contrast to the posterior hippocampal region of interest, no significant effects were found in the anterior hippocampal region of interest. A decrease in CMR_{glc} was expected there because the anterior hippocampus and the amygdala, both included in our region of interest, are strongly connected with the rhinal cortex, especially with the entorhinal and perirhinal cortices, respectively (Van Hoesen and Pandya, 1975; Suzuki and Amaral, 1990; Suzuki, 1996). In rats with extensive entorhinal lesions, in addition to the hippocampal glucose hypometabolism previously mentioned, transient amygdala hypometabolism has been reported (Beck *et al*., 1996). Despite the lack of statistically significant effect in this region of interest, a trend existed with consistently lower post-surgical CMR_{glc} values in the RH than in the SH group (see Table 2). Another factor that might explain these negative findings is the distribution of neurotoxic damage in the rhinal areas. Indeed, the anterior perirhinal cortex, which is the rhinal part most connected to the amygdala (Suzuki, 1996), was essentially spared, and the medial part of the entorhinal cortex, which provides the main projection to the rostral hippocampus (Witter and Amaral, 1991) was not consistently damaged since its medial ends were not targeted at surgery in order to avoid extra damage (see Method and Fig. 2).

Despite this issue of variable damage topography, our findings of significant and sustained CMR_{glc} effects on neocortical association and posterior hippocampal regions induced by rhinal lesions are globally in agreement with both the classic pattern of cerebral hypometabolism and the severe neuronal loss in the rhinal cortex that are characteristic of moderate-to-severe Alzheimer's disease, as well as with the anatomical connections of the rhinal cortex. Thus, the metabolic changes in Alzheimer's disease may be partly due to a disconnection, which is consistent with the hypothesis that neurofibrillary tangle concentration in the hippocampal–amygdala–entorhinal complex may lead to cortical metabolic dysfunction by a remote effect (Mielke *et al*., 1996). Most likely, however, other pathophysiological changes occurring in Alzheimer's disease and not reproduced in our model also contribute to the observed glucose hypometabolism. Additionally, the present results provide a rational explanation for the observed significant relationships between atrophy in the hippocampal region, including the entorhinal cortex, and temporoparietal hypometabolism (Meguro *et al*., 1997; Yamaguchi *et al*., 1997). Finally, our correlational approach indicates that damage to the perirhinal cortex is preferentially involved in at least some of the observed decreases in CMR_{glc}. However, additional studies are required to define precisely the contribution of each rhinal area to these remote metabolic effects.

Using the same experimental paradigm, but in contrast to the present results, neurotoxic lesions of the basal forebrain cholinergic structures in baboons were shown to have only marginal effects on cortical CMR_{glc}, in spite of severe cholinergic deafferentation (Le Mestric *et al*., 1998). In addition, declarative memory is severely impaired following lesions of the rhinal cortex (Gaffan and Murray, 1992; Meunier *et al*., 1993; Blaizot *et al*., 1997), but not of the cholinergic structures (Voytko *et al*., 1994). Taking both these and the present findings into consideration, neuronal loss in the rhinal cortex therefore appears to play a more important role in cerebral dysfunction in Alzheimer's disease than that involving the cholinergic system.

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